📖 Vue.js Complete Introduction Guide

A modern and beginner-friendly guide to building interactive web apps.

📄 What is Vue.js?

Vue.js is a **progressive JavaScript framework** used to build user interfaces and single-page applications (SPAs).

It focuses on:

- Simplicity ☺
- Flexibility ⚡
- Performance 🚀

In simple terms:

Vue = HTML + CSS + JavaScript + Reactivity ✨

🕰 History of Vue.js

Year	Event
2014	Vue.js released
Creator	Evan You
Inspired By	Angular & React

Vue quickly became popular for its **easy learning curve**.

⭐ Why Use Vue.js?

Key Benefits

✓ Easy to learn
✓ Lightweight and fast
✓ Reactive data binding
✓ Component-based architecture
✓ Strong community support

⚡ Core Features of Vue

1. Reactive Data Binding

UI updates automatically when data changes.

2. Components

Reusable UI building blocks.

3. Directives

Special HTML attributes like `v-if`, `v-for`.

4. Virtual DOM

Fast rendering and updates.

5. Single File Components (SFC)

HTML, CSS, and JS in one file.

🛠 Installing Vue

Using CDN

```
<script src="https://unpkg.com/vue@3"></script>
```

Using CLI

```
npm create vue@latest
```

💡 Basic Example

```
<div id="app">
  {{ message }}
</div>

<script>
const app = Vue.createApp({
  data() {
    return {
      message: "Hello Vue!"
    }
  }
});

app.mount('#app');
</script>
```

📦 Vue Data

```
data() {
  return {
    name: "John",
    age: 25
  }
}
```

🔧 Methods

```
methods: {
  greet() {
    alert("Hello!");
  }
}
```

🔄 Directives

Directive	Purpose
v-if	Conditional rendering
v-for	Looping
v-bind	Bind attributes
v-model	Two-way binding

Example:

```
<p v-if="isVisible">Hello</p>
```

□ Components

```
app.component('my-component', {
  template: `<h1>Hello Component</h1>`
});
```

🎁 Props

```
props: ['title']
```

Used to pass data between components.

🔄 Computed Properties

```
computed: {
  fullName() {
    return this.firstName + " " + this.lastName;
  }
}
```

🎧 Watchers

```
watch: {
  name(newValue) {
    console.log(newValue);
  }
}
```

⚛ Vue with Modern Tools

Common tools:

- Vue Router (routing)

- Pinia / Vuex (state management)
- Vite (build tool)

□ Example Program

```
<div id="app">
  <h1>{{ message }}</h1>
</div>

<script>
Vue.createApp({
  data() {
    return {
      message: "Hello Vue!"
    }
  }
}).mount('#app');
</script>
```

□ Advantages of Vue

✓ Simple and beginner-friendly
✓ Flexible architecture
✓ Fast performance
✓ Great documentation

● Disadvantages of Vue

• Smaller ecosystem than React
• Less corporate backing than Angular

🏆 Best Practices

✓ Use components properly
✓ Keep code modular
✓ Use Vue CLI / Vite

✓ Avoid large components
✓ Use state management for big apps

📌 Quick Summary

Vue.js helps developers:

- Build interactive UIs
- Create scalable applications
- Write clean frontend code

✨ Tips to Make Your PDF Look Professional

• Use green theme (Vue style) □
• Add UI screenshots
• Highlight code blocks
• Use fonts like:

- Inter
- Poppins
- Roboto

Recommended Layout:

- Cover Page
- Table of Contents
- Sections
- Code Examples
- Final Summary